

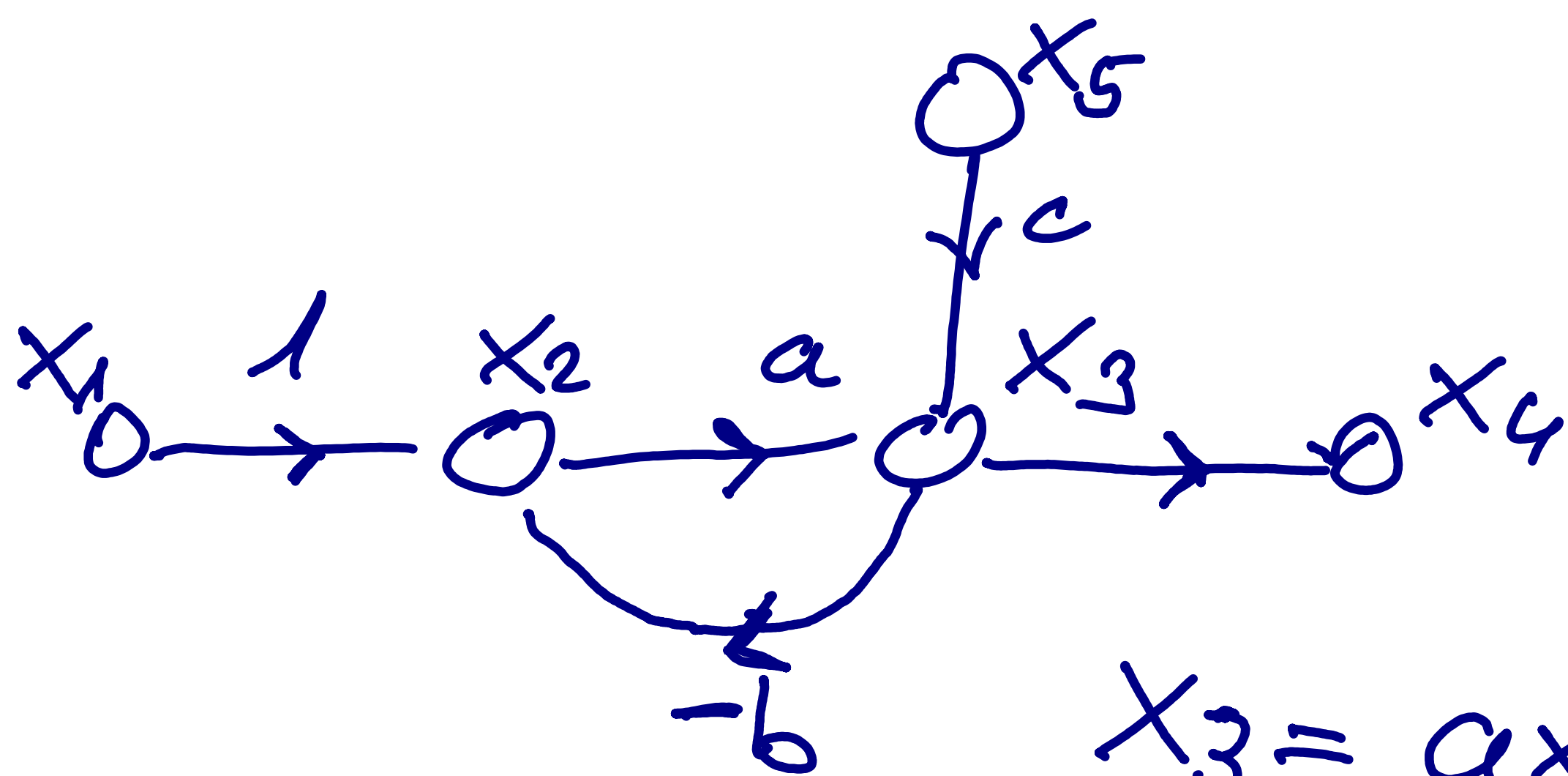
Graf de fluxuri

10.11.2025

Elementele componente ale unui graf de fluxuri sunt noduri și ramuri.

Noduri - variabilă sau semnal al sistemului

Ramuri - segment orientat de linie care conectează 2 noduri. Direcția de transmitere a semnalului este precizată cu o săgeată. În cazul sistemelor dinamice, fct. de transfer s.n. transmutante.



$$\begin{aligned}x_3 &= ax_2 + cx_5 \\x_2 &= 1x_1 - bx_3 \\x_4 &= 1x_3\end{aligned}$$

$x_1, \dots, x_5$ - Noduri

x_1, x_5 - Noduri de intrare

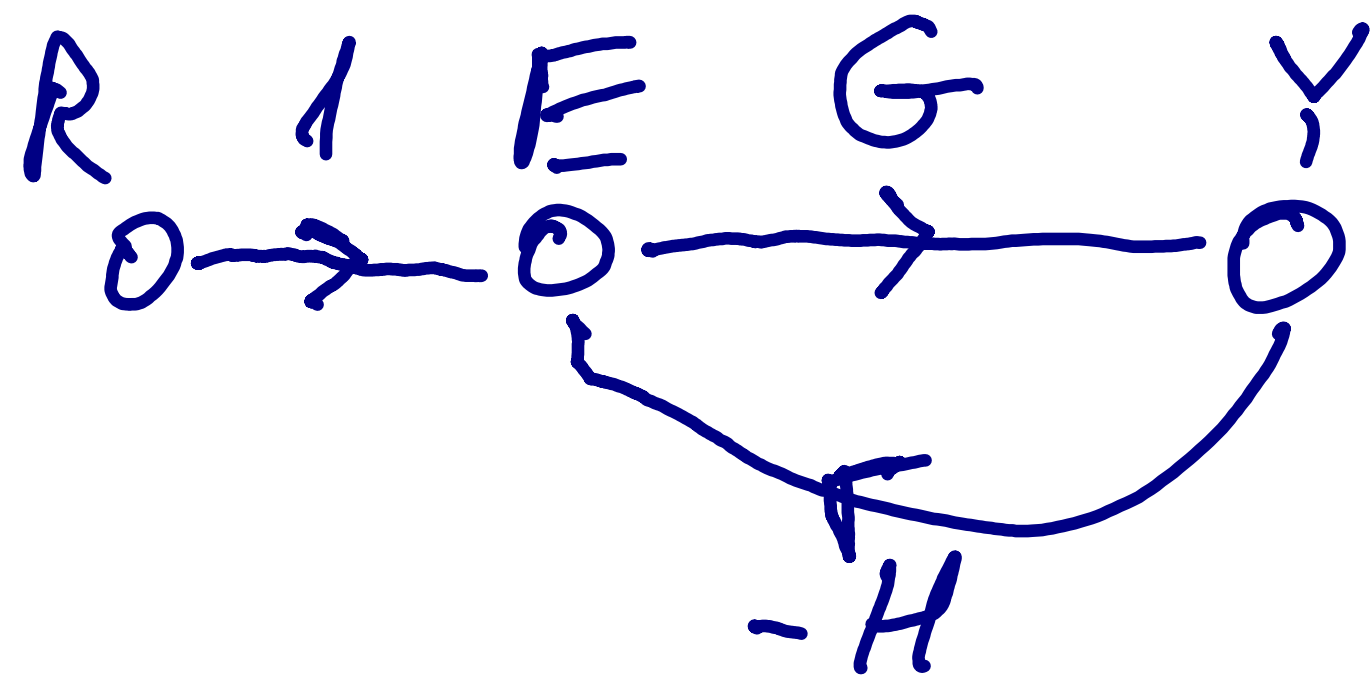
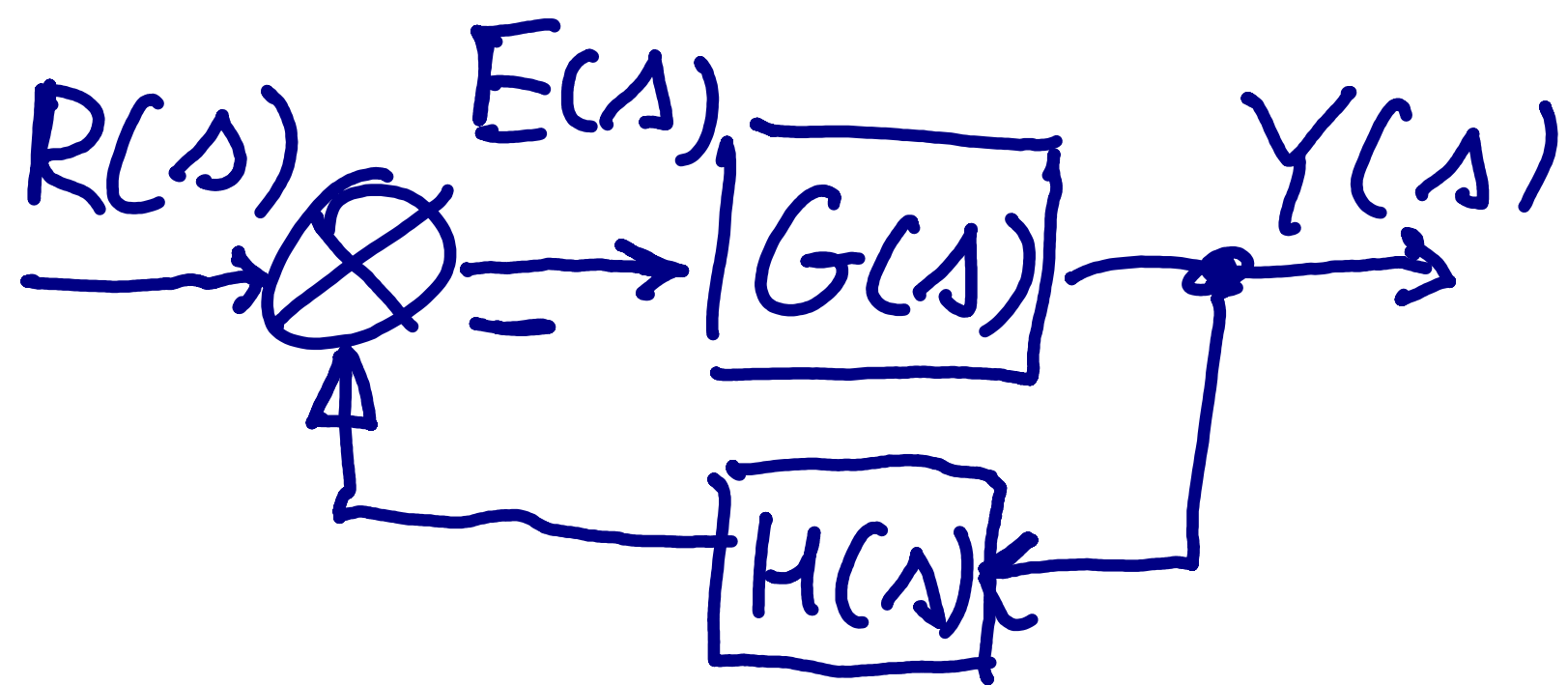
x_4 - Nod de ieșire

x_2, x_3 - Noduri mixte

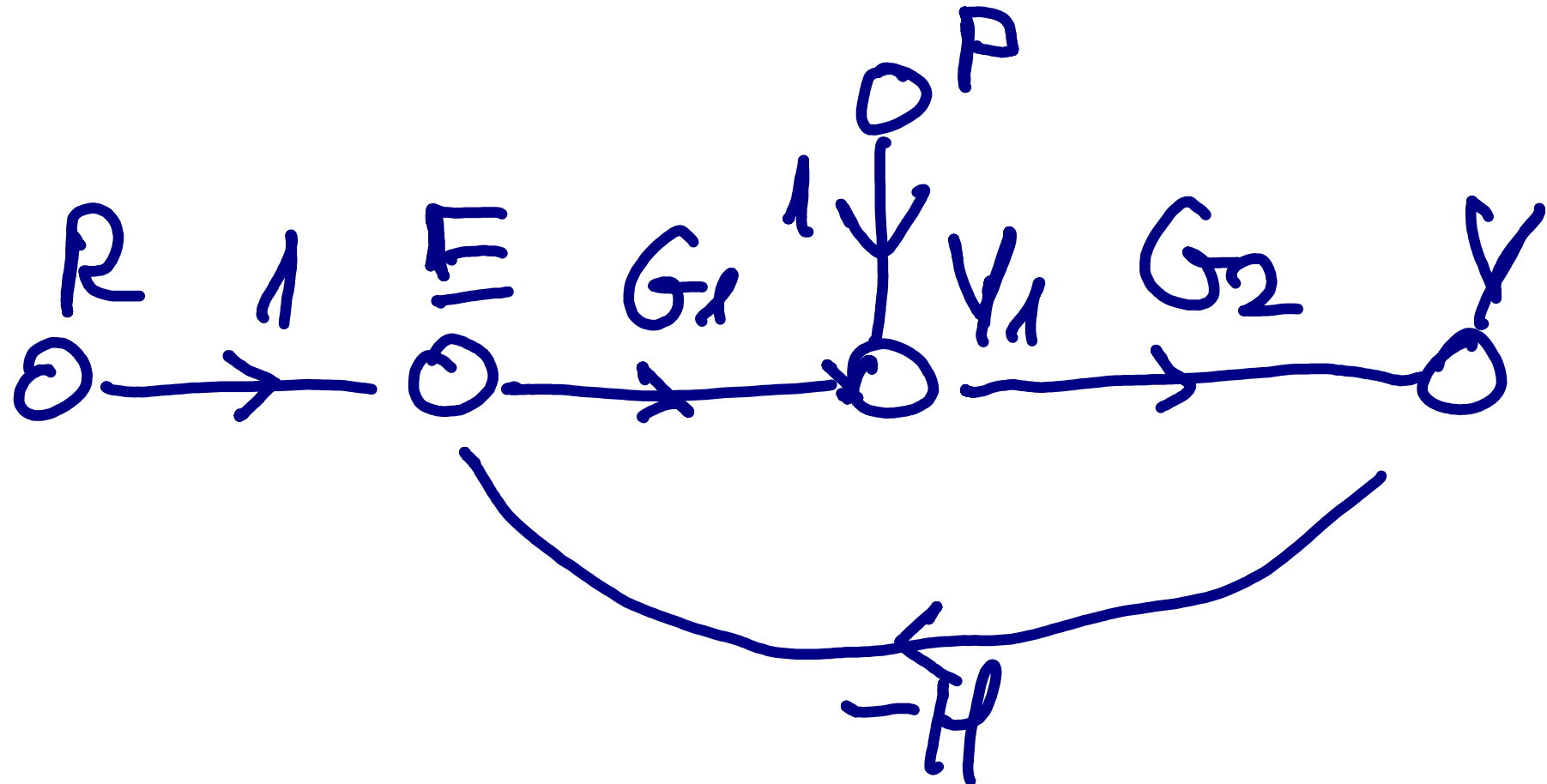
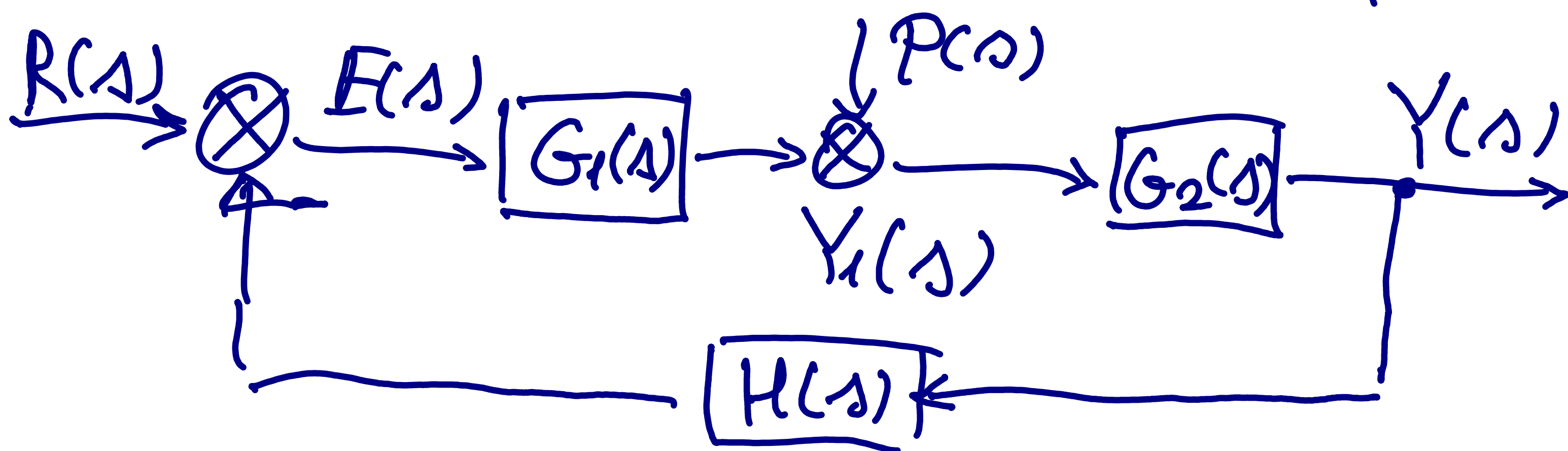
a, b, c - transmutante

- Adiacență : Două bucle sunt adiacente dacă au cel puțin un nod comun.

Modelarea sistemelor de reglare cu graf de fluentă :



Semnalele de intrare / ieșire, pt. de ramificație și pt. de sumare se reprezintă într-un graf de fluentă prin noduri. Nodurile sunt repr. prin intermediul ramurilor, iar transmitanța este egală cu funcțiile de transfer.



Formula lui Mason:

Pentru a determina transmitanța echivalentă sau f. de transfer echivalentă se folosește formula lui Mason.

$$G_e(s) = \frac{Y(s)}{R(s)} = \frac{1}{\Delta(s)} \sum_n G_{cdn}(s) \Delta_n(s)$$

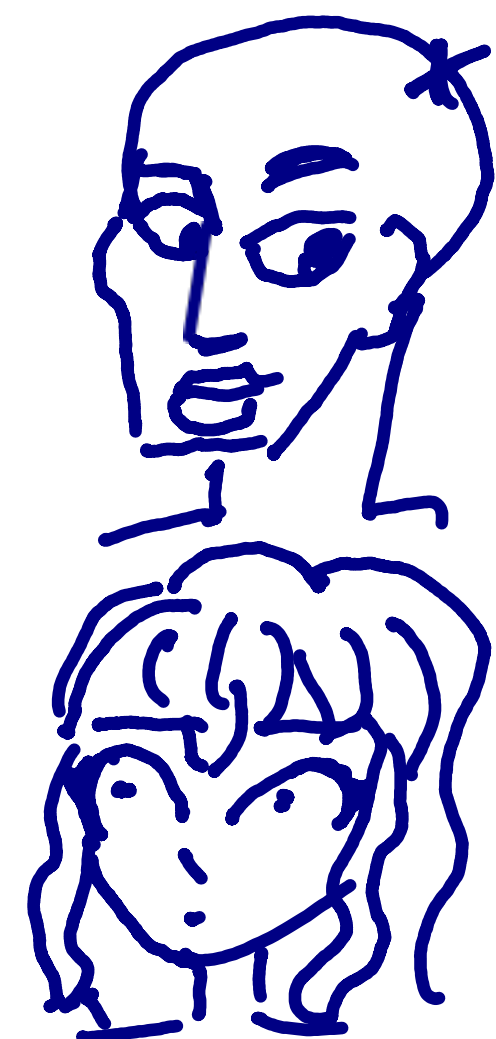
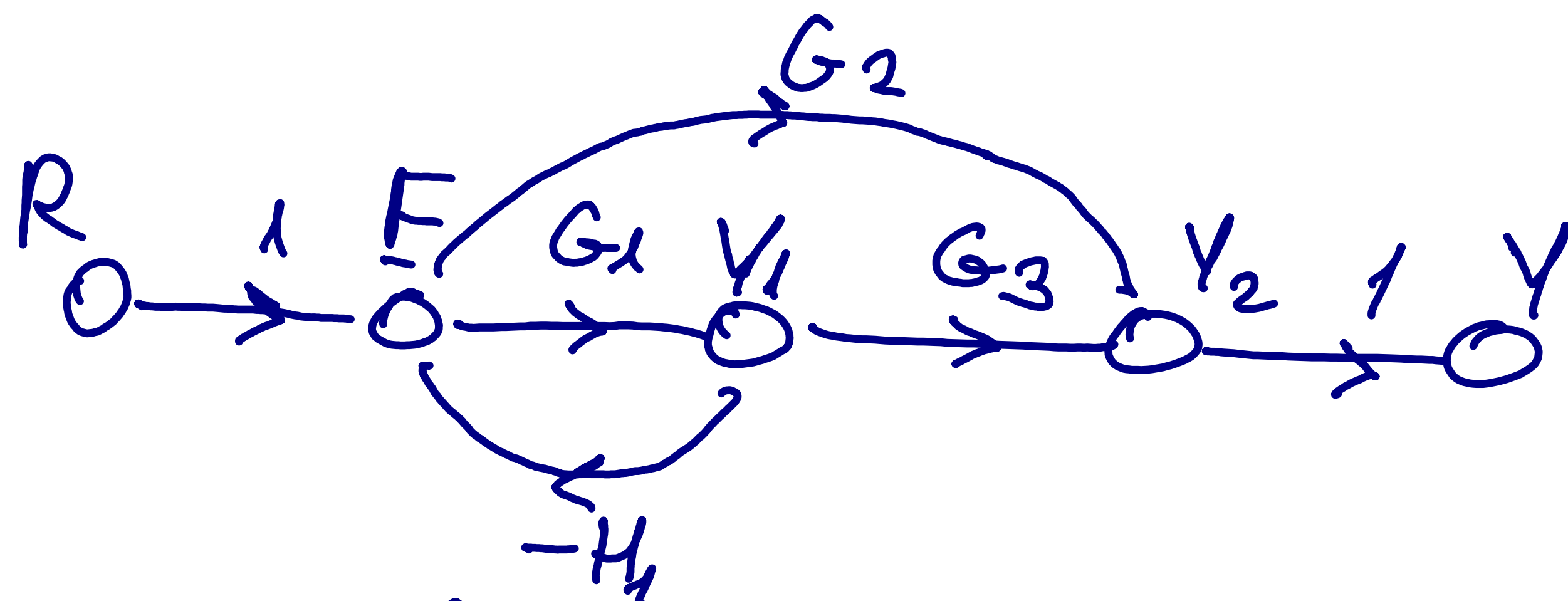
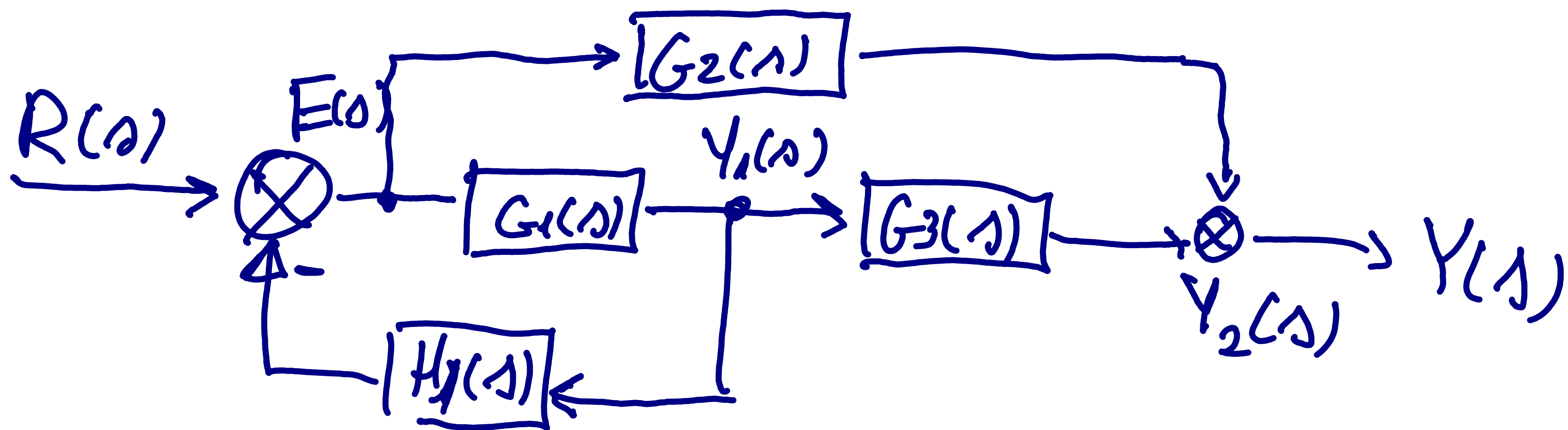
$\Delta(s)$ - determinantul grafului

$\Delta(s) = 1 - (\text{suma fct. de transfer a tuturor buclor individuale}) + (\text{suma produselor fct. de transfer a tuturor combinațiilor posibile de 2 bucle neadiacente}) - (\text{suma produselor fct. de transfer a tuturor comb. posibile de 3 bucle neadiacente}) + \dots - \dots +$

G_{cd} - funcția de transfer a căii directe

$\Delta_n(s)$ - factorul căii directe (din determinant se elimină toate buclele adiacente căii directe).

ex:



$$cd_1 : R \xrightarrow{1} E \xrightarrow{G_1} Y_1 \xrightarrow{G_3} Y_2 \xrightarrow{1} Y \quad G_{cd1} = G_1 G_3$$

$$cd_2 : R \xrightarrow{1} E \xrightarrow{G_2} Y_2 \xrightarrow{1} Y \quad G_{cd2} = G_2$$

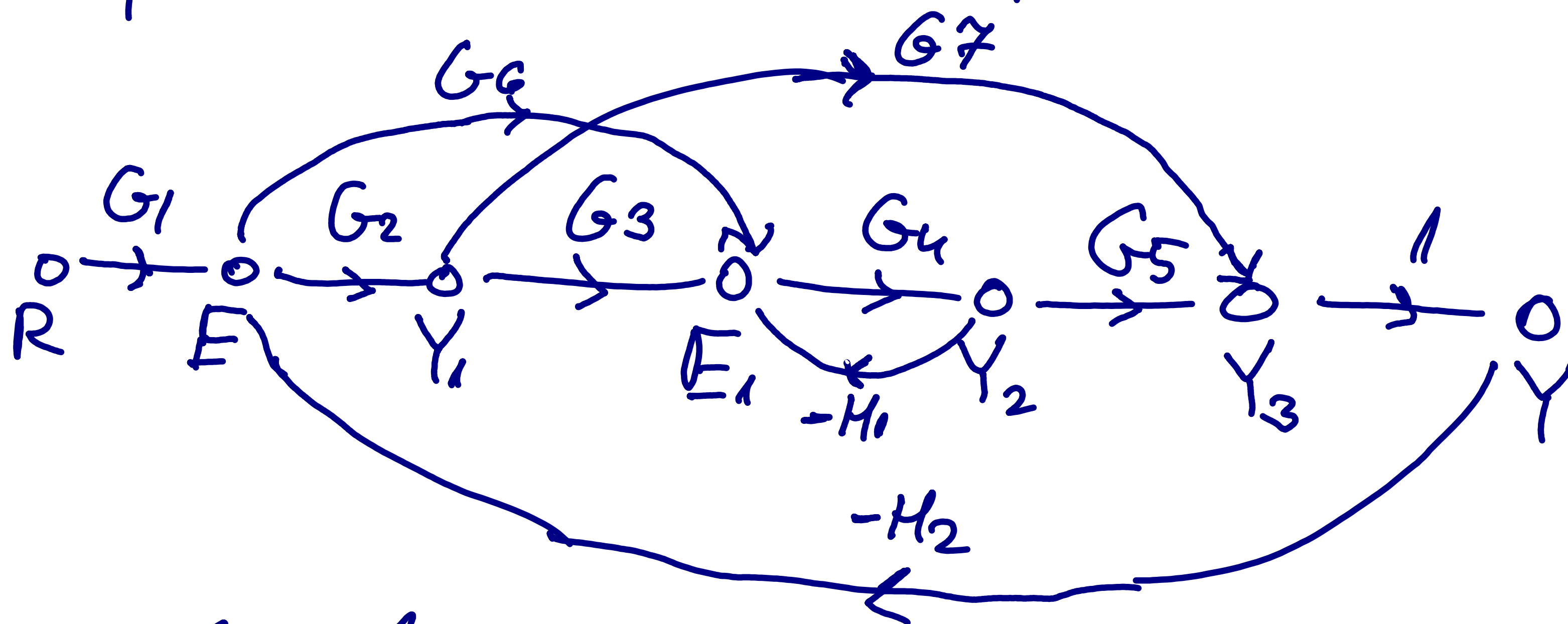
$$b_1 : E \xrightarrow{G_1} Y_1 \xrightarrow{-H_1} E \quad G_{b1} = -G_1 H_1$$

$$\Delta = 1 - G_{b1} = 1 + G_1 H_1$$

$$\Delta_{cd1} = 1$$

$$G_e = \frac{G_1 G_3 \cdot 1 + G_2 \cdot 1}{1 + G_1 H_1}$$

$$\Delta_{cd2} = 1$$



$$cd_1 : R \xrightarrow{G_1} E \xrightarrow{G_2} Y_1 \xrightarrow{G_3} E_1 \xrightarrow{G_4} Y_2 \xrightarrow{G_5} Y_3 \xrightarrow{1} Y, G_{cd1} = G_1 G_2 G_3 G_4 G_5$$

$$cd_2 : R \xrightarrow{G_1} E \xrightarrow{G_6} E_1 \xrightarrow{G_4} Y_2 \xrightarrow{G_5} Y_3 \xrightarrow{1} Y \quad G_{cd2} = G_1 G_6 G_4 G_5$$

$$cd_3 : R \xrightarrow{G_1} E \xrightarrow{G_2} Y_1 \xrightarrow{G_7} Y_3 \xrightarrow{1} Y \quad G_{cd3} = G_1 G_2 G_7$$

$$b_1 : E_1 \xrightarrow{G_4} Y_2 \xrightarrow{-H_1} E_1, G_{b1} = -G_4 H_1$$

$$b_2 : E \xrightarrow{G_2} Y_1 \xrightarrow{G_3} E_1 \xrightarrow{G_4} Y_2 \xrightarrow{G_5} Y_3 \xrightarrow{1} Y : G_{b2} = -G_2 G_3 G_4 G_5 H_2$$

$$b_3: E \xrightarrow{G_6} E, \xrightarrow{G_4} Y_2 \xrightarrow{G_5} Y_3 \xrightarrow{1} Y \xrightarrow{-H_2} E; \quad Gb_3 = -G_6G_4G_5H_2$$

$$b_4: Y_1 \xrightarrow{G_7} Y_3 \xrightarrow{1} Y \xrightarrow{-H_2} E \xrightarrow{G_2} Y_1; \quad Gb_4 = -G_7G_2H_2$$

$$\Delta = 1 - (Gb_1 + Gb_2 + Gb_3 + Gb_4) + Gb_1Gb_4$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1$$

$$\Delta_3 = 1 - Gb_1$$